

Michigan Tech

Background Math

Matrices & ODEs & parameters, Oh My!

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Summary

- It is fairly easy to numerically solve $y' = f(t, y) \quad y(t_0) = x$.
- Most solvers need the ODE in exactly this form with y a vector.
- Most solvers return vectors $y_1, y_2, \dots, y_N$ which approximate $y(t_1), y(t_2), \dots, y(t_N)$.
- Any higher order ODE can be written as a system.

Background Math: Probably Review

We need to recall and understand:

- How to solve constant-coefficient linear ODEs.
- How to solve constant-coefficient linear ODE systems.
- What a complex exponential means.

Background Math: Probably New

We need to learn or maybe recall:

- How to interpret eigenvalues and eigenvectors including complex eigenvalues.
- What the spectral decomposition $A = V\Lambda V^{-1}$ of a diagonalizable matrix A means.
- What the Singular Value Decomposition (SVD) $A = USV^T$ means.
- Why theorems are important.
- The definition of the matrix exponential e^A and what it is good for.

Background Computation:

We need to be able to:

- Numerically solve and plot the solution of a system of ODEs $y' = f(t, y)$
- Compute and use eigen values $\lambda_1, \dots, \lambda_n$ and vectors v_i of a matrix $A = V\Lambda V^{-1}$
- Compute and use singular values $s_1, \dots, s_n$ and vectors u_i and v_i of $A = USV^T$
- Reduce higher order ODEs to a first order system.

Solving Homogeneous Linear Constant Coefficient ODEs

Standard Form

$$ay'' + by' + cy = 0, \quad a \neq 0$$

Plugging the ansatz $y = e^{rt}$ gives the **characteristic equation**

$$ar^2 + br + c = 0.$$

Of course, this quadratic generically has two roots. Three Cases

1. Distinct real roots $r_1 \neq r_2$
2. Repeated real root $r_1 = r_2$
3. Complex conjugate roots $r = \alpha \pm i\beta$

Worked Example: Real

Solve $y'' - 5y' + 6y = 0$.

$$r^2 - 5r + 6 = 0$$

$$(r - 2)(r - 3) = 0$$

$$r = 2, 3$$

$$y(t) = C_1 e^{2t} + C_2 e^{3t}$$

Worked Example: Repeated

Solve $y'' + 4y' + 4y = 0$.

$$r^2 + 4r + 4 = 0$$

$$(r + 2)(r + 2) = 0$$

$$r = -2, -2$$

$$y(t) = C_1 e^{-2t} + C_2 t e^{-2t}$$

Worked Example: Complex

Solve $y'' + 4y' + 20y = 0$.

$$r^2 + 4r + 20 = 0$$

$$(r + 2)(r + 2) + 16 = 0$$

$$(r + 2)^2 = -16$$

$$r + 2 = \pm\sqrt{-16}$$

$$r = -2 \pm 4i$$

$$\begin{aligned} y(t) &= D_1 e^{(-2+2i)t} + D_2 e^{(-2-2i)t} \\ &= C_1 e^{-2t} \cos(4t) + C_2 e^{-2t} \sin(4t) \end{aligned}$$

Background Math: Maclaurin Series

$$\begin{aligned}
 \sum_{n=0}^{\infty} \frac{t^n}{n!} &= e^t = 1 + t + \frac{t^2}{2!} + \frac{t^3}{3!} + \frac{t^4}{4!} + \frac{t^5}{5!} + \frac{t^6}{6!} + \frac{t^7}{7!} + \frac{t^8}{8!} + \cdots, \\
 \sum_{n=0}^{\infty} (-1)^n \frac{t^{2n+1}}{(2n+1)!} &= \sin t = 0 + t + 0 - \frac{t^3}{3!} + 0 + \frac{t^5}{5!} + 0 - \frac{t^7}{7!} \cdots, \\
 &= \sum_{n=0}^{\infty} (-1)^n \frac{t^{2n}}{(2n)!} = \cos t = 1 + 0 - \frac{t^2}{2!} + 0 + \frac{t^4}{4!} + 0 - \frac{t^6}{6!} + 0 + \frac{t^8}{8!} + \cdots
 \end{aligned}$$

Complex Exponentials and Real ODE Solutions

For complex roots $\lambda = \alpha + i\beta$ the corresponding complex solution is

$$y(t) = e^{\lambda t} = e^{(\alpha + i\beta)t} = e^{\alpha t} e^{i\beta t}.$$

From the exponential power series

$$\begin{aligned} e^{i\beta t} &= 1 + i\beta t + \frac{(i\beta t)^2}{2!} + \frac{(i\beta t)^3}{3!} + \frac{(i\beta t)^4}{4!} + \dots \\ &= \left(1 - \frac{(\beta t)^2}{2!} + \frac{(\beta t)^4}{4!} - \dots\right) + i\left(\beta t - \frac{(\beta t)^3}{3!} + \frac{(\beta t)^5}{5!} - \dots\right) \\ &= \cos(\beta t) + i\sin(\beta t). \end{aligned}$$

Real solutions for real ODEs

So

$$e^{(\alpha+i\beta)t} = e^{\alpha t} \cos(\beta t) + i e^{\alpha t} \sin(\beta t)$$

$$e^{(\alpha-i\beta)t} = e^{\alpha t} \cos(\beta t) - i e^{\alpha t} \sin(\beta t)$$

The real and imaginary parts give two convenient real solutions for real ODEs

$$y_1(t) = e^{\alpha t} \cos(\beta t), \quad y_2(t) = e^{\alpha t} \sin(\beta t).$$

$\vec{y}' = A\vec{y}$: The Matrix Exponential e^A

For a square matrix A , the matrix exponential is defined by

$$e^A = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \frac{A^4}{4!} + \cdots = \sum_{k=0}^{\infty} \frac{A^k}{k!}$$

If $A = V\Lambda V^{-1}$ then

$$e^A = Ve^{\Lambda}V^{-1} = V \begin{pmatrix} e^{\lambda_1} & & 0 \\ & \ddots & \\ 0 & & e^{\lambda_n} \end{pmatrix} V^{-1}, \quad \Lambda = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$$

$Y' = AY$: Fundamental Solution $Y = e^{At}$

For A be an $n \times n$ constant matrix. The solution of the IVP

$$Y' = AY, \quad Y(0) = \mathbb{I}_n$$

is $Y = e^{At} = I + At + \frac{A^2 t^2}{2!} + \frac{A^3 t^3}{3!} + \cdots = \sum_{k=0}^{\infty} A^k \frac{t^k}{k!}$ since

$$\begin{aligned} Y'(t) &= \frac{d}{dt} \left[\sum_{k=0}^{\infty} A^k \frac{t^k}{k!} \right] = \sum_{k=0}^{\infty} \frac{d}{dt} \left[A^k \frac{t^k}{k!} \right] = \sum_{k=0}^{\infty} A A^{k-1} \frac{d}{dt} \left[\frac{t^k}{k!} \right] \\ &= A \sum_{k=0}^{\infty} A^{k-1} \left[\frac{k t^{k-1}}{k!} \right] = A \sum_{k=0}^{\infty} A^{k-1} \left[\frac{t^{k-1}}{(k-1)!} \right] = A \sum_{j=0}^{\infty} A^j \frac{t^j}{j!} = AY(t) \end{aligned}$$

The solution of $y' = Ay$ with $y(0) = y_0$ is $y = e^{At} y_0$

Computing e^{At} when $A = V\Lambda V^{-1}$

When $A = V\Lambda V^{-1}$

$$A^2 = V\Lambda V^{-1}V\Lambda V^{-1} = V\Lambda^2 V^{-1}$$

$$A^3 = V\Lambda^2 V^{-1}V\Lambda V^{-1} = V\Lambda^3 V^{-1}$$

$$\vdots$$

$$A^n = V\Lambda^n V^{-1}$$

For the spectral decomposition of $A = V\Lambda V^{-1}$ the

$$Y' = AY, \quad Y(0) = \mathbb{I}_n$$

is $Y = e^{At} = I + At + \frac{A^2 t^2}{2!} + \frac{A^3 t^3}{3!} + \dots = \sum_{k=0}^{\infty} A^k \frac{t^k}{k!}$ since

Solving $y' = Ay$ with $y(0) = y_0$

Substituting the *ansatz* $y = e^{\lambda t}v$ into $y' = Ay$ gives

$$\boxed{\frac{d}{dt} (e^{\lambda t}v) = A(e^{\lambda t}v)} \implies \boxed{\cancel{\lambda e^{\lambda t}v} = \cancel{e^{\lambda t}}Av} \implies \boxed{(A - \lambda \mathbb{I})v = 0}$$

The ansatz $y = e^{\lambda t}v$ solves the ODE iff (λ, v) is an eigen pair of A .

Superposition: Solution of $y' = Ay$, $y(0) = y_0$

If A has n LI evecs $V = (v_1 | \dots | v_n)$ with evals $\lambda_1, \dots, \lambda_n$ then

$$y = c_1 e^{\lambda_1 t} v_1 + \dots + c_n e^{\lambda_n t} v_n \quad \text{where} \quad V \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} = y_0$$

Summary

- It is fairly easy to numerically solve IVP.
- Most solvers need the function $f(t, y)$ for the ODE

$$y' = f(t, y)$$

- Most solvers return a list of vectors approximating $y_1, y_2, \dots, y_N$ the solution $y(t)$ at a sequence of points $t_1, t_2, \dots, t_N$.
- It is not hard to rewrite a higher order ODE as a system.